

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 244 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Variable Separable & Homogeneous DE	225 PYQs	Separable DE
Very High	Linear Differential Equations & Integrating Factor	47 PYQs	Linear DE

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Variable Separable & Homogeneous DE	$\int \frac{dy}{g(y)} = \int f(x) dx + C$, $\quad \text{For homogeneous: substitute } y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$
Linear Differential Equations & Integrating Factor	$\frac{dy}{dx} + P(x)y = Q(x) \implies \text{I.F.} = e^{\int P(x) dx}$, $\quad y \cdot \text{I.F.} = \int Q(x) \cdot \text{I.F.} dx + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Variable Separable & Homogeneous DE (Very High Yield — 225 PYQs)

Core Principle: Classification of order and degree, separation of variables, and reduction of homogeneous equations via $y = vx$.

■ **Caution / JEE Pitfall:** Degree of a differential equation is defined ONLY when the equation is a polynomial in its derivatives (no fractional powers or $\sin(dy/dx)$).

■ **Speed Trick:** Equations of the form $\frac{dy}{dx} = f(ax + by + c)$ can be solved by substituting $z = ax + by + c$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x\sqrt{1-x^2}dy + (y\sqrt{1-x^2} - x\cos^{-1}x)dx = 0, x \in (0,1)$, $\underset{x \text{ right...}}{\text{Find } y(1)}$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $(\frac{dy}{dx} + (1+x+x^2)(1-y+y^2), y(0)=1/2)$. Then $(2y(1)-1)$ is equal to..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x(\frac{dy}{dx} - y) = x^2 \cot x, x \in (0,\pi)$. If $y(\frac{\pi}{2}) = \frac{\pi}{2}$..." [Ans: N/A]

3.2 Linear Differential Equations & Integrating Factor (Very High Yield — 47 PYQs)

Core Principle: Standard first-order linear differential equations and integrating factor technique for $y(x)$ and $x(y)$.

■ **Caution / JEE Pitfall:** If the equation has $x \frac{dy}{dx}$, remember to divide by x throughout first so the coefficient of $\frac{dy}{dx}$ is 1!

■ **Speed Trick:** If the equation is non-linear in y , check if it is linear in x : $(\frac{dx}{dy}) + P(y)x = Q(y)$ implies I.F. = $e^{\int P(y)dy}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x(\frac{dy}{dx}) - y = x^2 \cot x$, $x \in (0, \pi)$. If $y(\frac{\pi}{2}) = \frac{\pi}{2}$..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let $f: R \rightarrow R$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in R$ and $f(0) \neq 0$. Let $g: [1, \infty) \rightarrow R$ be a differe..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x(\frac{dy}{dx}) - \sin 2y = x^3(2 - x^3)\cos^2 y$, $x \neq 0$. If $y(2) = 0$, then $\tan(y(1)) = \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let $y=y(x)$ be the solution of the differential equation $(\frac{dy}{dx}) = (1+x+x^2)(1-y+y^2)$, $y(0) = (1/2)$. Then $(2y(1)-1)$ is equal to

- (A) $\sqrt[3]{\tan(\frac{11\sqrt[3]{6}}{12})}$
- (B) $\frac{\sqrt[3]{2}\tan(\frac{11\sqrt[3]{12}}{12})}{\sqrt[3]{2}}$
- (C) $\sqrt[3]{\tan(\frac{11\sqrt[3]{12}}{12})}$ [Correct]
- (D) $\frac{\sqrt[3]{2}\tan(\frac{11\sqrt[3]{6}}{12})}{\sqrt[3]{2}}$

Correct Answer: C

Solution Analysis: Given $(\frac{dy}{dx}) = (1+x+x^2)(1-y+y^2)$, $y(0) = (1/2)$ We will solve it by the method of separation of variables. Since $\frac{dy}{1-y+y^2} = (1+x+x^2)dx$ Now simplify the denominator on the left: $1-y+y^2 = y^2 - y + 1 = (y - \frac{1}{2})^2 + \frac{3}{4}$ So, $\int \frac{dy}{(y - \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = \int (1+x+x^2)dx$ Using the standard result $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1}(\frac{u}{a})$ we get $\frac{2}{\sqrt{3}} \tan^{-1}(\frac{2y-1}{\sqrt{3}}) = x + \frac{x^2}{2} + \frac{x^3}{3} + C$ Now use the initial condition $y(0) = (1/2)$. Then $2y-1=0$ so $\$$

Benchmark Problem 2: JEE Main 2024 (Online) 30th January Evening Shift (Q#3)

Let $Y=Y(X)$ be a curve lying in the first quadrant such that the area enclosed by the line $Y-y=Y'(x)(X-x)$ and the co-ordinate axes, where (x,y) is any point on the curve, is always $\frac{y^2}{2Y'(x)+1}$, $Y'(x) \neq 0$. If $Y(1)=1$, then $12Y(2)$ equals

Correct Answer: 20

Solution Analysis: $A = \frac{1}{2} (\frac{yY'(x)+x}{Y'(x)})(y - xY'(x)) = \frac{y^2}{2Y'(x)+1} \Rightarrow (-y+xY'(x))(y-xY'(x)) = \frac{y^2}{2Y'(x)+1} - y^2 + xyY'(x) + xyY'(x) - x^2[Y'(x)]^2 = -y^2 + 2Y'(x) \frac{y^2}{2Y'(x)+1} + 2xy - x^2Y'(x) = 2$ & $\frac{dy}{dx} = \frac{2xy-2}{x^2} \parallel \& \frac{dy}{dx} - \frac{2}{x}y = \frac{-2}{x^2} \parallel \& \text{I.F.} = e^{\int -2\ln x dx} = \frac{1}{x^2} \parallel \& y \cdot \frac{1}{x^2} = \frac{2}{3}x^{-3} + c \parallel \& \text{Put } x=1, y=1 \parallel \& 1 = \frac{2}{3} + c \parallel \& c = \frac{1}{3} \parallel \& Y = \frac{2}{3}x^3 + \frac{1}{3}x^2 \parallel \&$